Modeling Toronto Apartment Rental Prices

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Introduction

This section should describe the background and objectives of the study.

Data Description

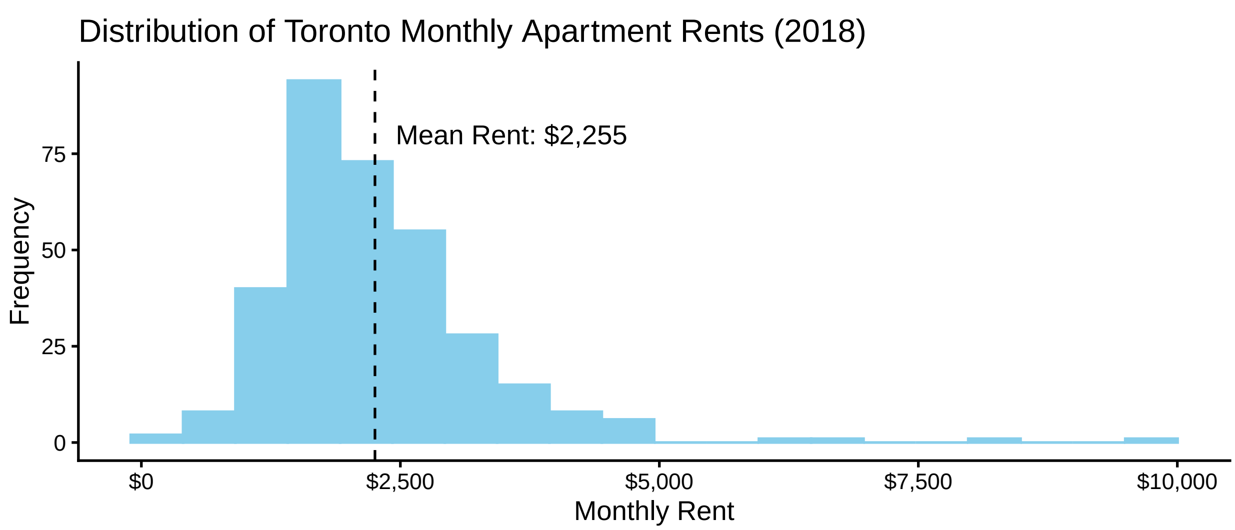
We obtained the data from this Kaggle dataset. The amount of rows in the original dataset before any data cleaning is 1,124. The data is made up of rental prices and contains data on apartment rental units in Toronto collected in 2018. It contains the apartment's geographic information (address, latitude, longitude), information on the rooms in each unit (the number of bedrooms, bathrooms, and dens), and the unit's monthly rental cost.

Exploratory Data Analysis

The total number of rows in the original dataset is 1,124. During data cleaning, we removed 308 duplicates which likely occurred because the data came from multiple sources. We also removed 5 rows with unusual values for `price` ($65/month, $99/month, $99/month, $36,900/month, $535,000/month). These outliers may skew our results so we decided to remove them. The sample size is now `n = 811`. There are no missing values in any columns.

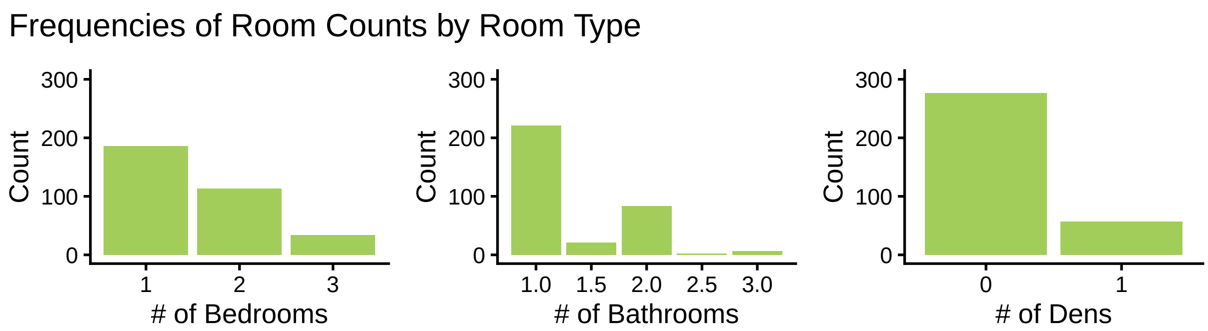
Before modeling, we want to find out more about the data. The variables we're most interested in are `price`, `bedroom`, `bathroom`, and `den`.

This histogram of `price` shows that the distribution is slightly right-skewed and unimodal around the mean of $2,255. As well, its standard deviation is $1,021, the minimum is $150, and the maximum is $9,750. TODO: more commentary



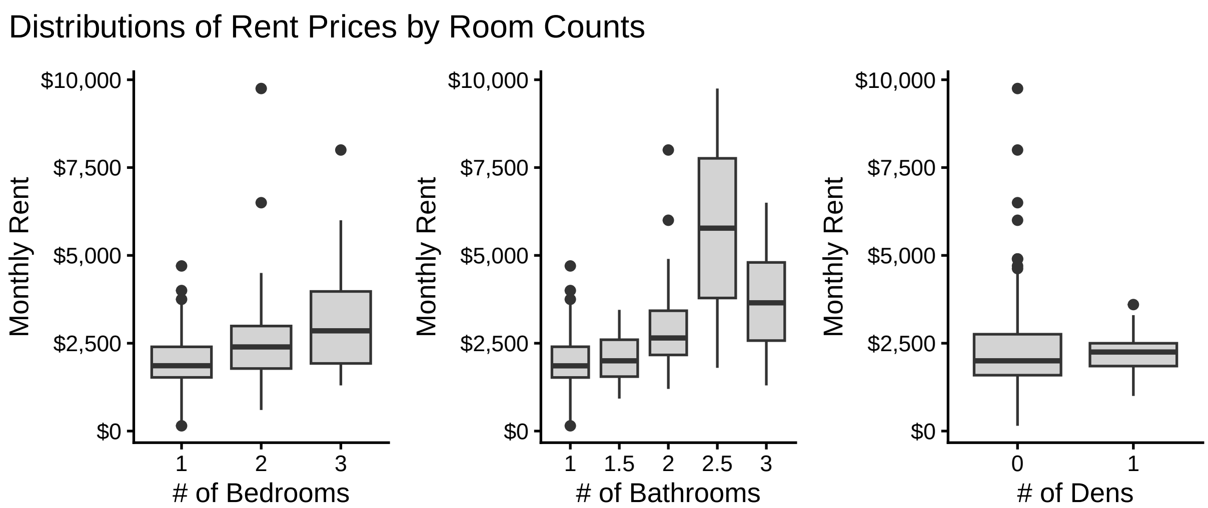
Next, we view the frequencies of each discrete value of `bedroom`, `bathroom`, and `den`.

TODO: comment on them (such as the unbalanced groups)

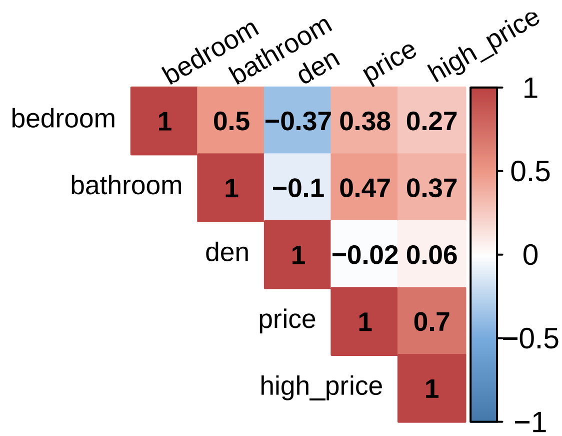


Below are boxplots of `price` split by `bedroom`, `bathroom`, and `den` counts in order to see how `price`'s distribution varies by each of these 3 apartment characteristics.

TODO: comment on them



We're also curious about the correlations between the variables so we plot (the upper triangle of) a correlation matrix.



The variables with the strongest correlation with `price` are `bathroom` (r = 0.47) and `bathroom` (r = 0.38), though these are only moderately strong linear correlations. `den` is weakly negatively correlated with `price` at r = -0.01. Moreoever, `bedroom` and `bathroom` are moderately correlated at r = 0.50, which may indicate that they're collinear, which may affect our modeling decisions later on.

TODO: key take-aways of EDA

Models

This section should include at least two statistical models for the same dataset, justifications of the choices of the models, model diagnostics, parameter estimates and their standard errors and p-values, and interpretation of the results.

```
##
## Call:
## lm(formula = price ~ ., data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1374.3  -355.0   -42.2   295.0  6258.0
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   923.02     124.39   7.420 1.01e-12 ***
## bedroom       195.38      70.05   2.789  0.00559 **
## bathroom      373.82      91.16   4.101 5.20e-05 ***
## den            25.18     108.69   0.232  0.81696
## high_price   1243.62      82.70  15.037  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 682.6 on 328 degrees of freedom
## Multiple R-squared:  0.5583, Adjusted R-squared:  0.5529
## F-statistic: 103.6 on 4 and 328 DF,  p-value: < 2.2e-16
```

```
##
## Call:
## lm(formula = price ~ bedroom + bathroom, data = data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2653.8  -542.1   -48.6   452.4  6470.0
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)    794.5      149.5   5.315 1.97e-07 ***
## bedroom        294.6       83.8   3.515 0.000501 ***
## bathroom       758.6      114.0   6.654 1.19e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 889.8 on 330 degrees of freedom
## Multiple R-squared:  0.2448, Adjusted R-squared:  0.2403
## F-statistic:  53.5 on 2 and 330 DF,  p-value: < 2.2e-16
```

Conclusion

This section should provide overall conclusions, any limitations in the datasets and analyses, and validity of the conclusions.

Appendix

```r
library(tidyverse)
```